

Lean port sheets

Rev F0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | D-044 lean ports A (PV) and B (battery) + insulation monitor, BUS- referenced

01 Supplies and commands

Live 24 V / 5 V / 3.3 V, BUS- tie, commands

File: sheets/01_lean_io.kicad_sch

02 Port A power path (pv)

Varistor Y + monitor

File: sheets/02_leanA_power.kicad_sch

03 Port A sensing and interlocks

References, BUS- shunt amplifier + hold-off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/03_leanA_ctrl.kicad_sch

04 Port A coil drives

Contactor: interlock logic, fail-safe high side, economiser; passive bank bleeder

File: sheets/04_leanA_drive.kicad_sch

05 Port B power path (battery)

Varistor Y + monitor, aR fuses

File: sheets/05_leanB_power.kicad_sch

06 Port B sensing and interlocks

References, BUS- shunt amplifier + hold-off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/06_leanB_ctrl.kicad_sch

07 Port B coil drives

Contactor + precharge relay: interlock logic, fail-safe high side, economiser; passive bank bleeder

File: sheets/07_leanB_drive.kicad_sch

08 Insulation monitor

PE divider, two switched 992 k strings through CA-IS3417WT

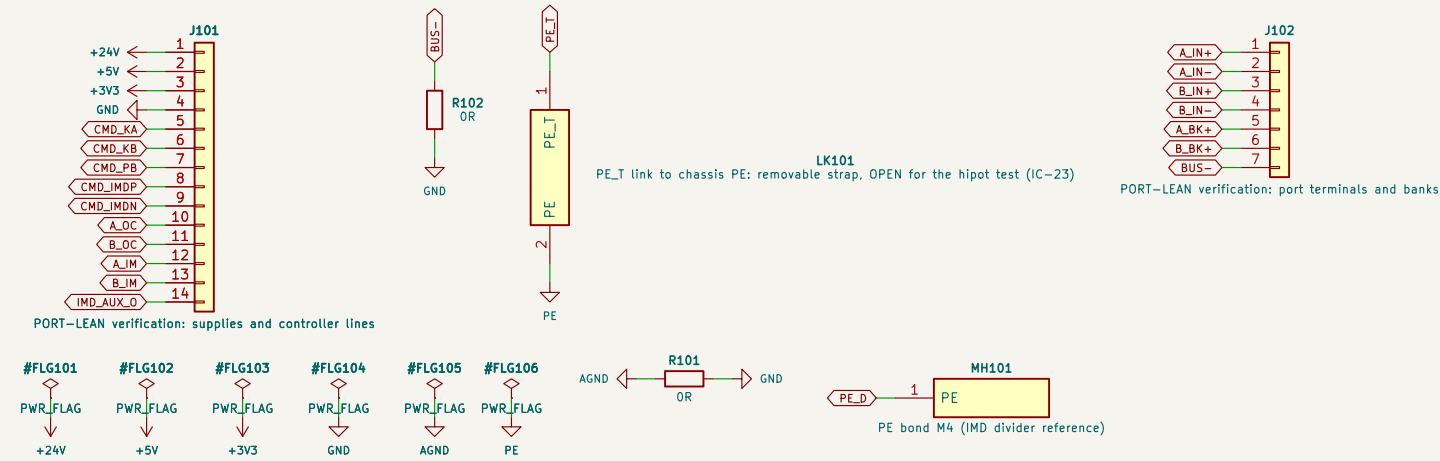
File: sheets/08_lean_imd.kicad_sch

Verification build of the reusable lean-port functions of gen/port.py (lean_port, varistor_y, lean_contactor, lean_precharge, lean_bleeder, lean_dividers, lean_shunt, lean_interlock, lean_imd). The product sheets live on the power board.

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 1
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: / File: PORT-LEAN.kicad_sch		
Title: Lean port sheets – Cover / Hierarchy		
Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 1/9

SUPPLIES AND COMMANDS



On the power board these come from the AUX-T1 live winding and the controller.

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 2
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /01 Supplies and commands/
File: 01_lean_io.kicad_sch

Title: Lean port sheets – Supplies and commands

Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 2/9

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal slide A_{LN} / A_{JN} - bank A_{JK}+. No fuse, no precharger; array loc. <= 1.25 x rating.
 Contactor opens <= 0.97 sA, held above (port spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A/JN+ / A/JN-, bank A/BK+. No fuse, no precharge; array Isc <= 1.25 x rating.
 Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

TV25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE₃ Up, eff 3793 V pole-P and pole-pole at In 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA. Branch fuses ETI CH4x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CN765B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 4
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /03 Port A sensing and interlocks/
File: 03_leanA_ctrl.kicad_sch

Title: Lean port sheets – Port A sensing and interlocks

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 4/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 5
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /04 Port A coil drives/
File: 04_leanA_drive.kicad_sch

Title: Lean port sheets – Port A coil drives

Size: A3

Date: 2026-10-04

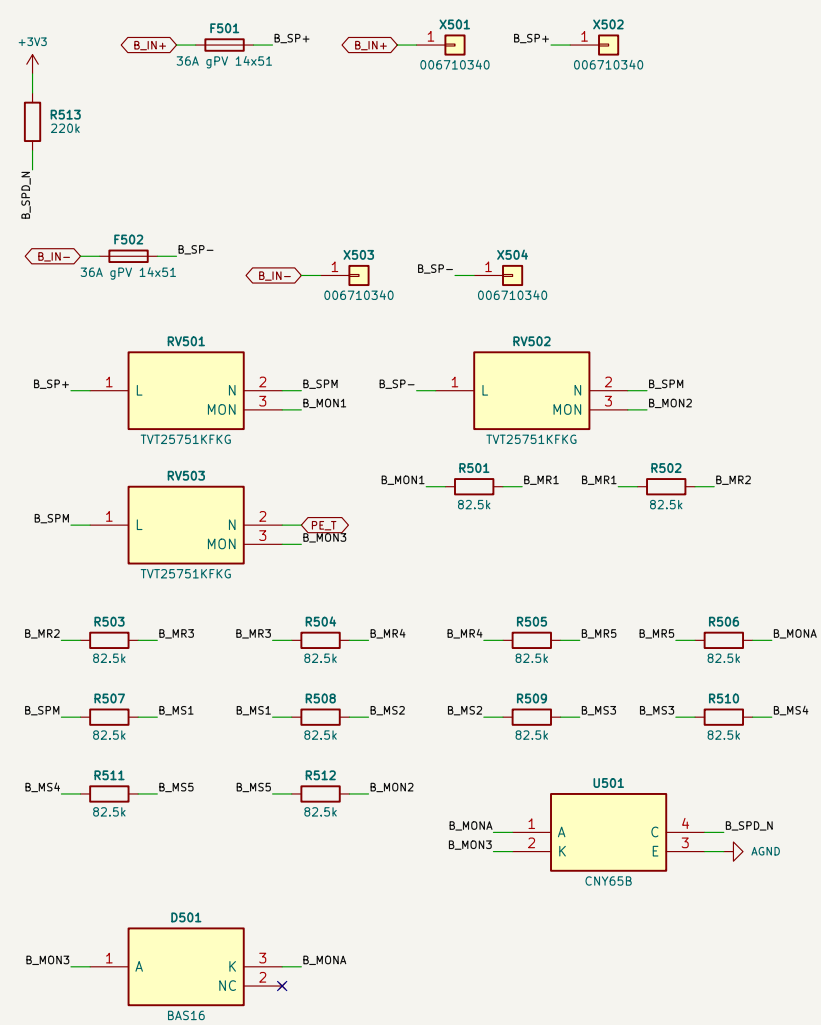
Rev: F0

KiCad E.D.A. 10.0.6

Id: 5/9

Terminal side B_{UN}+ / B_{UN}-, bank B_{BK}-. HPE501 250 A aR in both poles (breaks 1.25–50 kA only).
 Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

Terminal side B_{JN+} / B_{JN-}, bank B_{BK+}, HPIE501 250 A aR in both poles (breaks 1.25–50 kA only).
Circuit breaker opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

[illegible]

TV125751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE_T. Up,eff 3793 V pole-PE and pole-pole at 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); Imax 25 kA, Branch fuses ETI CH1x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B

The schematic diagram illustrates a multi-channel signal processing system. It begins with a series of input buffers (U601-U609) that receive signals from various sources. These buffers are followed by comparators (U701-U709) which compare the input signals against reference voltages. The outputs of the comparators are then processed by logic gates (U801-U809) to generate the final digital outputs. The circuit also includes a variety of passive components, such as resistors (R1-R100) and capacitors (C1-C10), which are used for signal conditioning and timing. The components are interconnected in a complex network, with each component's pins and connections clearly labeled. The diagram is organized into several functional blocks, each with its own set of pins and connections. The components are represented by standard electronic symbols, and the connections are shown as lines with labels indicating the signal names and pin numbers. The overall layout is a detailed representation of the physical circuit, showing the flow of signals from input to output through various processing stages.

HPE501/010B100-250 in both poles: breaks 1.25-50 kA only (partial range)

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 7
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /06 Port B sensing and interlocks/
File: 06_leanB_ctrl.kicad_sch

Title: Lean port sheets – Port B sensing and interlocks

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 7/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'
Sheet 8
Schematics authored by Chinmoy Bhuyan
Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /07 Port B coil drives/
File: 07_leanB_drive.kicad_sch

Title: Lean port sheets – Port B coil drives

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 8/9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values: sim/port_design.py -> port_spec.json 'lean'

Sheet 9

Schematics authored by Chinmoy Bhuyan

Live side referenced to BUS-; basic HV-PE; functional inside the live side

Sheet: /08 Insulation monitor/

File: 08_lean_imd.kicad_sch

Title: Lean port sheets – Insulation monitor

Size: A3

Date: 2026-10-04

Rev: F0

KiCad E.D.A. 10.0.6

Id: 9/9